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Low-dissipation tunable rf preamplifier for low temperature NMR applications

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We describe a simple circuit which can be used as an impedance matching device to improve the S/N ratio when lossy cables must be used in low-temperature NMR experiments. The total dissipation is less than 2 mW and the circuit operates satisfactorily down to 1.5 K. The only tuned circuit is the nuclear resonance circuit which incorporates GaAs (P) diodes which can be tuned remotely.

INTRODUCTION

NMR experiments at low temperatures often necessitate the separation of the NMR coil from a room temperature operated spectrometer by a long length (typically 1 m) of coaxial cable. A rf current passes through the coaxial line and the losses are high. For experiments at very low temperatures (e.g., those obtained by dilution refrigeration) one must minimize the thermal conductivity of the cable in order to reduce the heat load on the refrigerator and the rf losses become very important. A considerable improvement in the signal-to-noise ratio can therefore be expected if one can use a preamplifier operating at helium temperatures, either close to the NMR coil, or such that a short length of low-loss low-temperature cable (e.g., a superconducting cable¹) can be used between the amplifier and the coil.

Although some low-temperature circuits have been described for this purpose they have been limited to special cases. Such circuits include a twin-T bridge operating essentially at a fixed frequency,² marginal oscillators³⁻⁷ which cover a limited frequency range (and whose rf levels are difficult to reduce to levels suitable for cw work at low temperatures), and the cascode circuit of Miyoshi and Cotts⁸ designed specifically for pulsed NMR. The latter should in principle be suitable for cw experiments but it suffers from the drawback that it dissipates approximately 100 mW, which is excessive for a 4 K helium bath and prohibitive for a small pumped liquid helium bath. We sought a more general approach to the problem and report here a simple circuit operating at helium temperatures which uses a minimum number of components and which can be operated over a wide frequency range. The total dissipation is less than 2 mW and the amplifier can be used in conjunction with a pumped helium liquid bath.

I. GENERAL CONSIDERATIONS

In the elementary circuit of Fig. 1(a) where the NMR coil (impedance Z_L) is separated from the room-temperature spectrometer by a length l of coaxial cable

having a characteristic impedance Z_0 , the effective shunt impedance at the spectrometer input is

$$Z_T = \frac{Z_L + Z_0 \tanh(\alpha + j\beta)l}{1 + (Z_L/Z_0) \tanh(\alpha + j\beta)l}.$$

α and β are the attenuation and propagation factors of the cable. If one uses half-wavelength line lengths (or multiples thereof) in order to maximize the positive reactance at the spectrometer input, $\tanh(\alpha + j\beta)l \approx \alpha l$ for a low-loss line and

$$Z_T \approx \frac{Z_L + \alpha Z_0}{1 + (Z_L/Z_0)\alpha l}. \quad (1)$$

The additional series resistive loading αZ_0 is a problem even if low-loss cables can be used, $\alpha \approx 20$ dB/100 m at 30 MHz for typical small diameter cables (e.g., type RG-17U, 2.54 mm o.d.). These commercial cables are not suitable for very low temperature applications and for experiments using dilution refrigeration we have used

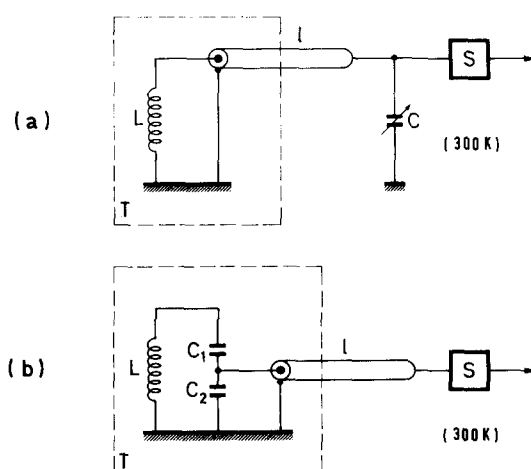


FIG. 1. Circuit configurations used for low temperature NMR studies. In (a) the nuclear resonance coil at low temperatures T is separated from the room-temperature spectrometer S by a length l of coaxial cable and L is resonated by the capacitance C at room temperatures. The rf losses due to the cable in configuration (a) can be reduced by using the circuit in (b) where the coil is resonated at low temperatures and impedance matched to the coaxial line by the capacitor divider C_1/C_2 .

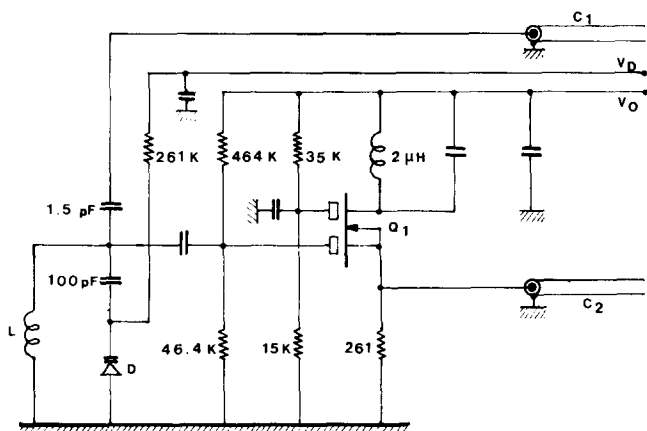


FIG. 2. Circuit diagram of the source follower. Q_1 is a MOS-FET RCA type 3N187, D is a GaAs(P) L.E.D. Monsanto type MV5024 which is tuned remotely by means of the voltage V_D . C_1 and C_2 are homemade 50- Ω low-temperature cables as described in the text. All capacitors are metal film types and those unmarked are nominally 100 nF. The resistors are also of the metal film type and their resistances were measured at 4 K (The nominal values are given in the circuit diagram).

small homemade 50- Ω cables using stainless steel or cupronickel outer conductors (3 mm o.d.), brass or cupronickel inner conductors, and PTFE tubing as the dielectric. The losses were estimated to be 40 dB/100 m at 30 MHz and for the circuit of Fig. 1(a) the effective series resistance $\alpha/Z_0 \approx 25 \Omega$, and the effective Q is almost entirely determined by the cable losses.

The problem is not resolved by putting the resonating capacitance C in the low temperature circuit [Z_L in Eq. (1) is real and large and the effective shunt loading α/Z_0 remains]. The losses can be reduced considerably, however, if the nuclear resonance circuit is impedance matched to the cable using two capacitances as shown in Fig. 1(b). If the fractional error in impedance matching is f , $Z_L = Z_0(1 + f)$, the input impedance at the spectrometer is $Z_T \approx (1 + f - 2\alpha/f)Z_0$, and the losses are considerably reduced for small f . Unfortunately, the capacitor divider of Fig. 1(b) requires two nonorthogonal adjustments and although reverse biased GaAs(P) diodes can be used as voltage variable tuning capacitors at helium temperatures,⁶ a simple broadband source follower circuit provides a more practical means of impedance transformation especially if one wishes to minimize the admixture of the nuclear dispersion signal.

II. PRACTICAL CIRCUIT

The circuit shown in Fig. 2 uses a MOS-FET source follower operating at 4 K to transform the impedance of the nuclear resonance circuit to a 50- Ω low-temperature coaxial cable. The resonance circuit is tuned using a reverse biased GaAs(P) light emitting diode: Monsanto type MV5024, $I_R \leq 200$ nA for $-20 < V_R < -3$ V. The capacitance varied from 9 pF for a reverse bias of -20 V to 35 pF for a bias of -1 V. [For this capacitance ratio one would need a cumbersome number of diodes in order to resonate and impedance match

TABLE I. Transconductance (in mmho) of some MOS-FETs for different temperatures.

Type	Temperature (K)			Operating bias	
	300	77	4.2	V_{gs} (V)	I_D (mA)
3N187	10	13	14	1.5	3
3N200	12	10	11	2.5	2
40673	9	14	8.5	1.5	4

a high Q coil to a 50- Ω cable using the circuit of Fig. 1(b).]

The characteristics of several MOS-FET's were explored and the results of the most promising are summarized in Table I. We chose the RCA type 3N187 because of its high transconductance at 4 K, the large gain-bandwidth feature (high g_m/C_0 ratio), and the stability of the I - V characteristics. Anomalous results were observed for some transistors for elevated values of the drain current and this was attributed to a heating of the substrate to temperatures above the cryogenic bath. For this reason provision must be made to anchor the transistor to a suitable heat sink if the circuit is not immersed in liquid helium. The circuit was operated at a drain current $I_D = 0.3$ mA with $V_{DS} = 3$ V in order to reduce the total dissipation to less than 2 mW. For these conditions the amplifier could be used attached to a small pumped helium bath (such as those in dilution refrigerators⁹) without excessive overloading. We tested the circuit while immersed in a pumped liquid helium bath and the amplifier worked satisfactory down to 1.5 K.

The noise figure of the circuit can be determined in terms of the equivalent voltage and current noise sources for the field effect transistor¹⁰ and one finds

$$F \approx 1 + \frac{2}{3}(1 + R_S G)^2 / g_m R_S + (4/15)(\omega/\omega_G)^2 g_m R_S.$$

G is the input admittance of the FET (approximately 1.2 mmho at 10 MHz for the 3N187), R_S is the resistance of the tuned resonance circuit, g_m is the transconductance of the FET, and $\omega_c = g_m/C_0$, where C_0 is the transistor's input capacitance. At 10 MHz the above expression gives a noise figure of 1.8 dB for $R_S = 20$ K Ω and this is to be compared with an observed noise figure of 2.9 dB.

III. PERFORMANCE

In order to compare the performance of the source follower circuit with the conventional arrangement of Fig. 1(a) we adopted a Q -meter configuration using two low-temperature cables, one to apply an rf current

TABLE II. Observed S/N ratios for the EPR absorption signal of 1,1-diphenyl-2-picrylhydrazyl at 17 MHz.

Circuit arrangement	Temperature (K)		
	300	77	4
Without preamplifier [Fig. 1(a)]	<1	2.5	33
Using source follower (Fig. 2)	10	58	230

to the coil via a nominal 1.5 pF metal film capacitance, and a second to couple the output of the preamplifier to a low-noise broadband rf amplifier operating at room temperature. (For the latter we used a wide band integrated circuit Robinson oscillator¹¹ with the feedback loop open.) The resonance signals following diode detection (bandwidth ≈ 100 KHz) were displayed on an oscilloscope and the S/N ratios of (a) an EPR absorption signal (≈ 10 mm³ 1,1-diphenyl-2-picrylhydrazyl), and (b) a ¹⁹F NMR absorption signal (≈ 10 mm³ PTFE) were studied at three different temperatures—300, 77, and 4.2 K. The results are summarized in Table II.

The improvement in signal to noise at 4 K was approximately 7 and although this is only slightly better than the improvement reported by Aldermann² for a fixed frequency circuit, the circuit presented here has greater flexibility and can be tuned over a significant frequency range, $f_{\max} \approx 2f_{\min}$ for a given NMR coil.

We also studied the performance of a common source amplifier using a tuned π -circuit to impedance match the output to the cable. For the same operating conditions this led to poorer results for the S/N ratio. The common source circuit showed gain variations for small variations of the V_{DS} voltage and this was to be expected since the transconductance is temperature dependent. For a given gain $A(\delta A/\delta T)_{C.S.} \approx A[(1/g_m)(\partial g_m/\partial T)]$. Self-

heating of the substrate can then lead to fluctuations in A if the transistor is not adequately heat sunked. The corresponding variations for the source follower, $(\partial A/\partial T)_{S.F.} \approx -(1/g_m R_0)(1/g_m)(\partial g_m/\partial T)$ are much smaller for $g_m R_0 \gg 1$. The simpler broadband source follower impedance matching circuit is therefore to be preferred over the narrow band common source amplifier circuit.

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